晶圓廠 KOYO POLYIMIDE 烘烤爐管 O.I.

KOYO POLYIMIDE 烘烤爐管 O.I.

壹、目的 (Purpose)：

一、製程目的：

Process purpose：

對 POLYIMIDE 之烘烤，請根據AUTOERS system所顯示程式run貨。

Polyimide curing.

貳、適用範圍 (Scope)：

一、設備/Machine：

KOYO LINDBERG VF-5100B

二、站別/Operation：

B-29 (F31)、B-48 (F43)

參、作業內容 (Contents)：

一、操作步驟/Operating steps：

晶舟座轉盤基本說明及上貨步驟：

Description of cassette turntable and loading steps:

1.1晶舟座轉盤 (TURN TABLE) 最多可放置八個晶舟，依序為 A、B、C、D、E、F、G、H，每個晶舟 (白色鐵弗龍) 上皆有標示其所屬晶舟轉盤之位置(A、B、C、D、E、F、G、H)，勿更動其位置；目前設定 TURN TABLE 上固定放置 7 個晶舟(A、B、C、D、E、F、G)，晶舟內皆擺滿 25 片檔片。

The cassette turntable can have at most 8 cassettes which are numbered in sequence A, B, C, D, E, F, G and H. Each cassette (white Teflon) has a marking on it to indicate its position (A, B, C, D, E, F, G or H) on the turntable which should not be altered; currently the default setting of the turntable is have 7 cassettes (A, B, C, D, E, F, and G) with each cassette fully loaded with 25 dummy wafers.

1.1 目前每次最多限 RUN 6 批，除 G 晶舟固定擺放檔片不動外；請依照實際產品之批數，

對照下表，以裝有產品之鐵弗龍晶舟接調換原 TURN TABLE 之晶舟，換下之晶舟連同其內之檔片暫放綠色晶舟盒內，盒上亦標示其原屬 TURN TABLE 之位置 (例如：C)， 但因每批產品實際不足 25 片，故不滿處，先以原先換下晶舟之檔片抽出補滿產品晶舟之空格，才將其放置 TURN TABLE 之對應位置上。

Currently at most 6 lots can be run at a time, except for G cassette which regularly has dummy wafers in it; to proceed, replace the original turntable cassettes with the Teflon cassettes loaded with products according to the actual number of lots to be run and in reference to the table below. The replaced cassettes along with the dummy wafers in them are placed temporarily in the green wafer box which is also marked with their original position in the turntable (e.g. C). For lots less than 25 pcs, make up with dummy wafers in the replaced cassette up to 25 pcs each cassette before placing the cassette in its corresponding position in the turntable.

例如：欲 RUN 產品一批 23 片，先將 C 位置之晶舟取出，抽出其中 2 片檔片補於產品晶舟內，再將產品晶舟放至原空出之 C 位置，其餘晶舟及其內檔片勿動。

Ex.: If you want to run a lot of 23 pcs, first take out the cassette at C position, draw 2 pcs of dummy wafers and put them in the product cassette, then place the product cassette in the now-empty C position, while leaving the other cassettes and the dummy wafers as is.

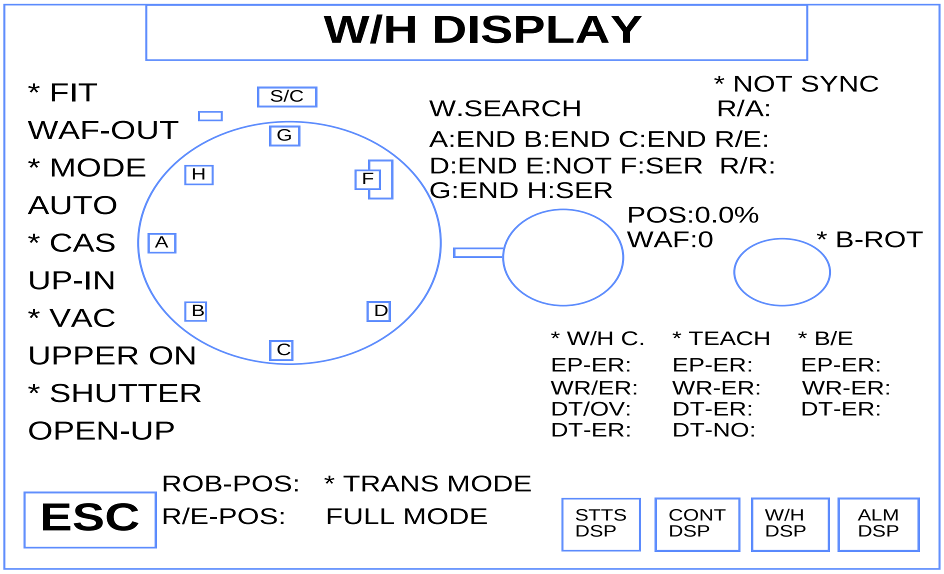
RUN 貨批數 Lots to be run	放入之晶舟位置 Cassette position
1	C
2	B、C
3	B、C、D
4	B、C、D、E
5	A、B、C、D、E
6	A、B、C、D、E、F

注意：RUN 貨請使用白色鐵弗龍晶舟。

Note: Use white Teflon cassette when running the products.

1.3 按螢幕右下方之 $\boxed{\text{W/H DSP}}$ 鍵，則進入晶舟座轉盤之控制螢幕，若需將 C 晶舟面向操作者，於螢幕上按 C 處，待晶舟轉向操作者，取出上貨 (或下貨)，其餘晶舟依同理操作。

Press the $\boxed{\text{W/H DSP}}$ key at the lower right corner of the screen to enter the control menu of the cassette turntable. If you want to turn Cassette C toward the operator, press C on the menu. After Cassette C faces the operator, remove it to load the products (or unload). Use the same steps for operate other cassettes.



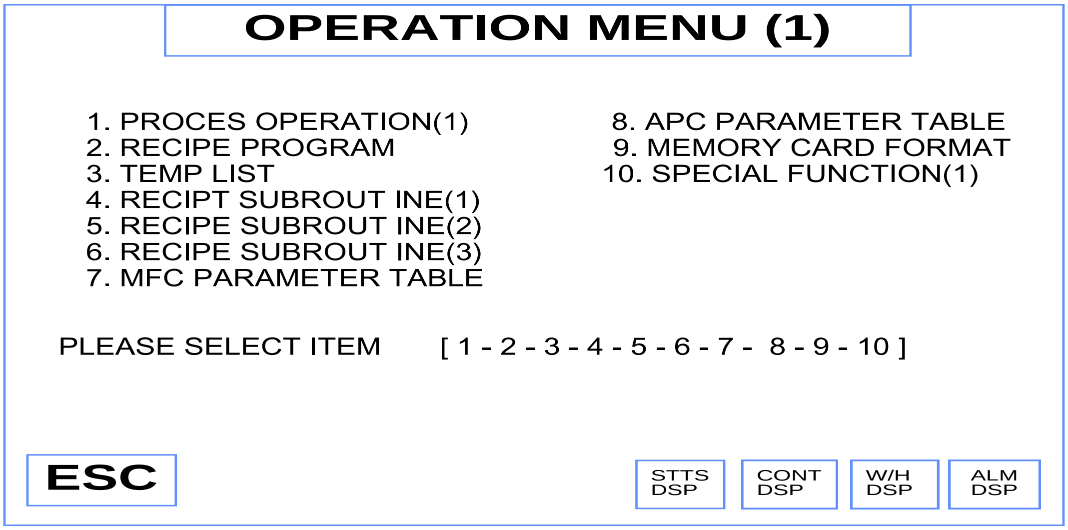
1.4 需確定所有晶片之平邊朝外並對齊。

Make sure the flat side of all wafers face outward and are aligned.

２.RUN 程式步驟(Run steps)：

2.1 重覆按控制面板 MODEL 1000 螢幕左下方之 ESC 鍵，直到其回到主螢幕 OPERATION MENU(1)，如下圖。

Press the ESC button at the lower left screen of control panel Model 1000 until the menu goes back to the main menu- the Operation Menu (1) as shown below:



2.2 於 OPERATION MENU(1) 之項目選擇按 "1" ，

Select Item "1" under the Operation Menu (1).

PLEASE SELECT ITEM [1-2-3-4-5-6-7-8-9-10]

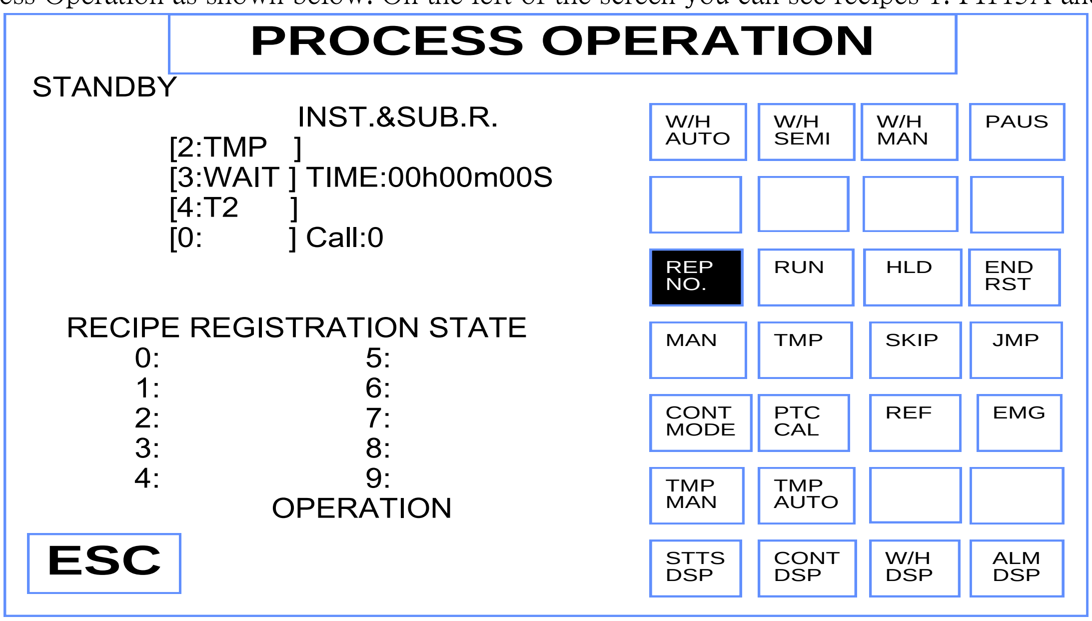
↑

按此鍵

Press this key to

則進入程式操作螢幕 PROCESS OPERATION，如下圖，螢幕左方可見現有程式 1:PI115A 及 6:O2BAKE。

enter the Process Operation as shown below. On the left of the screen you can see recipes 1. PI115A and 6:02BAke.



2.3 於 PROCESS OPERATION 螢幕上，輸入欲執行之程式，步驟如下：

Key in the recipe to be executed under the Process Operation by following the steps as follows:

2.3.1 按 [REP NO] 鍵，則出現數字幕。

Press [REP NO] and figures appear.

2.3.2 於數字幕按入欲執行之程式號碼，目前為程式 "1" (PI115A)。

Key in the No. of recipe to be executed. Currently it's "1" (PI115A).

2.3.3 螢幕出現 [YES-NO]，按 YES，稍待一會可見螢幕中下方出現 [SAVING!!!] 字樣，之後程式 0 出現 PI115A，顯示已將程式 1 (PI115A) 存入程式 0 內待執行。

When the screen shows [YES-NO}, press Yes, and the wording [SAVING!!!] appears on the bottom the screen a moment later, followed by PI115A under recipe 0, indicating recipe 1 (PI115A) has been stored into recipe 0 to be executed.

2.4 按 RUN 鍵，螢幕左下方出現 [RUN TO BE EXCUTED?] ，同時中下方出現 [YES-NO]，按 YES，則執行 2.3 所存入之程式，此時面板右上方 RUN 之方形綠燈會亮。

Press the Run key, [RUN TO BE EXECUTED?] appears on the left bottom of the screen and [YES-NO} in the mid-bottom. Press YES, the recipe saved in Step 2.3 is being executed and the square-shaped green indicator of run at the upper right corner of the panel will light.

3.下貨步驟 (Unload steps)：

3.1 當程式執行完至 END 時，面板右上方之 END 紅燈會亮，此時檢查晶片是否已 UNLOAD 回晶舟內 , 若 OK 至 PROCESS OPERATION 螢幕上按 [END RST] 鍵，則螢幕左下方出現 [END TO BE RESET?]，螢幕中下方出現 [YES-NO]，按 YES，則 END 之紅燈熄滅，RUN 之綠燈亦熄滅。

When the execution ends, the red END indicator at the upper right of panel will light. At this time, check whether the wafers have been unloaded back to the cassette. If OK, press the [END RST] button under Process Operation, and {END TO BE RESET?] appears on the lower left of the screen, and [YES-NO} appears in the mid-bottom. Press Yes, then the red END indicator will go off, and the green Run indicator also goes off.

3.2 按螢幕右下之 [W/H DSP] 鍵，進入晶舟座轉盤之控制螢幕，依序將原先上貨之晶舟取下，將補充之檔片取出放回原先之晶舟內，將原晶舟放回轉盤所屬位置上，重覆直到所有產品取出為止。

Press the [W/H DSP] button on the lower right of the screen to enter the control menu of the cassette turntable. Take out the cassettes by the original order of loading, then take out the dummy wafers and put them back into the original cassettes, and put these cassettes back to their corresponding positions in the turntable. Repeat the aforesaid steps until all products are taken out.

二、製程參數/Process parameter

1.程式內容/recipe contents：

1.1.RECIPE NUMBER：001

RECIPE NAME：PI115A F31(B29)

LINE	INST	VALUE 1	VALUE 2
1	CASCT		
2	PIDM	0	
3	T1		
4	TEMP	140	0
5	WAIT	0h0m15s	
6	CTO		
7	CTI		
8	BED		
9	SHC		
10	WAIT	0h0m15s	
11	LD		
12	SHO		
13	BEU		
14	PIDM	5	
15	WAIT	0h0m3s	
16	TMP	165	3
17	WAIT	0h5m0s	
18	PIDM	0	
19	WAIT	0h0m3s	
20	T2		
21	O2CHK	20m30s	0
22	TMP	400	3
23	WAIT	0h30m0s	
24	PIDM	1	

LINE	INST	VALUE 1	VALUE 2
25	WAIT	0h0m3s	
26	TMP	256	3.0
27	WAIT	0h5m0s	
28	V14	ON	
29	WAIT	0h0m40s	
30	T2		
31	WAIT	0h3m0s	
32	V14	ON	
33	WAIT	0h0m40s	
34	T2		
35	WAIT	0h3m0s	
36	TMP	140	3.5
37	T6		
38	WAIT	0h0m3s	
39	PIDM	0	
40	T1		
41	BED		
42	SHC		
43	WAIT	0h10m0s	
44	ULD		
45	CTO		
46	SHO		
47	BEU		
48	END		

1.2.RECIPE NUMBER：001

RECIPE NAME：PI115A F43(B48)

LINE	INST	VALUE 1	VALUE 2
1	CASCT		
2	PIDM	0	
3	T1		
4	TEMP	140	0
5	WAIT	0h0m15s	
6	CTO		
7	CTI		
8	BED		
9	SHC		
10	WAIT	0h0m15s	
11	LD		
12	SHO		
13	BEU		
14	PIDM	0	
15	WAIT	0h0m3s	
16	TMP	165	3
17	WAIT	0h5m0s	
18	T2		
19	O2CHK	0h20m0s	0
20	WAIT	0h0m1s	
21	PIDM	3	
22	TMP	400	3
23	WAIT	0h0m1s	
24	PIDM	8	
25	WAIT	0h5m1s	

LINE	INST	VALUE 1	VALUE 2
26	PIDM	1	
27	WAIT	0h25m0s	
28	TMP	256	3.0
29	WAIT	0h5m0s	
30	V14	ON	
31	WAIT	0h0m40s	
32	T2		
33	WAIT	0h3m0s	
34	V14	ON	
35	WAIT	0h0m20s	
36	T2		
37	WAIT	0h3m0s	
38	TMP	140	3.0
39	PIDM	2	
40	T6		
41	WAIT	0h0m3s	
42	PIDM	0	
43	T1		
44	BED		
45	SHC		
46	WAIT	0h10m0s	
47	ULD		
48	CTO		
49	SHO		
50	BEU		
51	END		

1.3. RECIPE NUMBER：006

RECIPE NAME：O2 BAKE(B48、B29)

LINE	INST	VALUE 1	VALUE 2
1	CAST		
2	PIDM	0	
3	T1		
4	TMP	140	0
5	T2		
6	WAIT		
7	PIDM	3	
8	TMP	750	12
9	WAIT	1 sec	
10	PIDM	1	

LINE	INST	VALUE 1	VALUE 2
11	WAIT	30 min	
12	PIDM	2	
13	TMP	135	8
14	T7		
15	WAIT	5 min	
16	PIDM	0	
17	T1		
18	TMP	140	0
19	WAIT	1 sec	
20	END		

2.W/H CONDITION：

1	WAFER	CHARGE 125 WAFERS DUMMY UPPER: 15 LOWER: 10 MONITOR 000-000-000-000-000-000-000 000-000-000-000-000-000-000 000-000-000-000-000-000
2	EMPTY IN	NOT EXECUTION
3	CASSETTE CONDITION	A: P B: P C: P D: P E: P F: N G: N H: N TYPE: 8 CAS. PIT: 4.760 mm
4	FINGER	TYPE: 5F-2C FINGER
5	BOAT.B/E	SLIT: 175 PIT: 4.762 mm APCAL.

3.溫控參數/Temperature control parameter：

3.1 B29

CASCADE OUTPUT LIMIT	TOP	50℃
	CENT	30℃
	BTOM	50℃

BCASCADE BIAS	TOP	-80℃
	CENT	-70℃
	BTOM	-50℃

3.2 B48

3.2.1 PID DATA

	TOP	CENT	BOTM
＋P [%]	3.5	3.5	5.0
＋I [m]	4.0	4.0	6.0
＋D [m]	0.3	0.5	0.5
－P [%]	3.5	3.5	5.0
－I [m]	4.0	4.0	6.0
－D [m]	0.3	0.5	0.5

3.2.2 CASCADE CONT PARAMETER

(0)	CASCADE OUTPUT LIMIT	TOP CENT BOTM	50.0 ℃ 50.0 ℃ 50.0 ℃
	CASCADE BIAS	TOP CENT BOTM	－50.0 ℃ －50.0 ℃ －50.0 ℃

(1)	CASCADE OUTPUT LIMIT	TOP CENT BOTM	50.0 ℃ 50.0 ℃ 50.0 ℃
	CASCADE BIAS	TOP CENT BOTM	－50.0 ℃ －50.0 ℃ －60.0 ℃

(2)	CASCADE OUTPUT LIMIT	TOP CENT BOTM	50.0 ℃ 50.0 ℃ 50.0 ℃
	CASCADE BIAS	TOP CENT BOTM	－250.0 ℃ －250.0 ℃ －250.0 ℃

(3)	CASCADE OUTPUT LIMIT	TOP CENT BOTM	50.0 ℃ 50.0 ℃ 50.0 ℃
	CASCADE BIAS	TOP CENT BOTM	－50.0 ℃ －60.0 ℃ －50.0 ℃

(8)	CASCADE OUTPUT LIMIT	TOP CENT BOTM	50.0 ℃ 50.0 ℃ 50.0 ℃
	CASCADE BIAS	TOP CENT BOTM	－40.0 ℃ －55.0 ℃ －60.0 ℃

此參數，負責之製程工程師可對應 FURNACE EXHAUST 狀況作調整以控制品質，調整範圍為±20℃之內。

To control quality PE can adjust cascade cont parameter by furnace exhaust condition ,adjusting range between ±20℃.

肆、操作注意事項 (Important Operation Notes)：

1.RUN 產品部份：

Product run:

上貨請確實依照壹之四操作步驟之規定，將產品換置於規定之晶舟內。

When loading, follow the instructions specified in I.D. Operating Steps faithfully to transfer the products in specified cassette.

2.機台部份：

The machine:

2.1 RUN貨中，除負責之製程工程師為達品質控制可自行微調製程參數，嚴禁其它人員更改調整程式或機台參數。

Except for responsible PE who may fine tune the process parameters for the purpose of quality control, anybody else is strictly prohibited to adjust the recipe or machine parameters during the run.

2.2 RUN 貨前須確認機台正常且在 STANDBY 狀態，請檢查：

Before starting the run, verify that the machine is normal and in standby mode and check the following:

2.2.1 MODEL 1000 控制面板左上方之氣閥配置面板上，代表閥 1、4 及 6 之黃燈為亮著。

On the gas valve panel at the upper left hand of Model 1000 control panel, the yellow lights representing valves 1, 4 and 6 are illuminated.

2.2.2 按 MODEL 1000 面板螢幕右下方 $\boxed{\text{STTS DSP}}$ 鍵，則顯示機台狀態，檢查 N2 流量正常 (PN2：10.0SLM) ，並檢查 STANDBY 溫度為正常 (PV：140.0±5.0℃)。

Press the $\boxed{\text{STTS DSP}}$ button at the lower right of the Model 1000 panel screen to show the machine status; check whether N2 flow is normal (PN2:10.0SLM) and check whether the Standby temperature is normal (PV: 140.0±5.0°C).

```
                STATUS DISPLAY
[READY]
                                                        R.TM:  0h 0m 0s
 TUBE No.0001                                           T.TM:  0h 0m 0s
 STANDBY      LNo:  0        INST:                      I.TM:  0h 0m 0s
 INTERLOCK  : No: 00         INST:          TM:  0h 0m 0s
[TEMP]   TEMP:140.0¢J   RP: 0.0¢J/m    C-MODE:CASCNT     T-MODE:AUTO
STADBY        [  TOP  ]   [ CENTR ]    [  BTM  ]
           SP:  140.0¢J      140.0¢J      140.0¢J
           PV:  139.9¢J      140.1¢J      140.0¢J

          VALVE:12345678901234       SEQ:1234567890123456789012 34
                 .   .  .      .

[VALVE   [ NAME ]  [  SET  ] [ MEASURE ] [ NAME ]  [  SET  ] [ MEASURE ]
  /GAS]   1: PN2        10.0    10.0 SLM    5: O2AN     0.0     0.0 SLM
NORMAL    2: PO2         0.0     0.0 SLM    6: APC     10.0    40.0 SLM
          3: BN2         0.0     0.0 SLM    7: HOLD     0.0     0.0 SLM
          4: PS1         0.0     0.0 SLM    8: CLOSE    0.0     0.0 SLM

[VAC     -APC-           0.0     0.0 Torr
 GAUGE]  -PIRANI1-             0.000 Torr   -Ex.PR.GAUGE-       41.2 mm/H2O
         -OXY.ANAL.-            52.3 ppm    -PIRANI4-          0.000 Torr

ESC                                    STTS   CONT   W/H    ALM
                                       DSP    DSP    DSP    DSP
```

2.3 晶片傳送時，若發生異常，請在面板上將 [W/H PAUSE] 鍵，按一次即傳送暫停，並立刻通知設備工程師處理。

If irregularity occurs during wafer transfer, press the [W/H PAUSE] button on the panel once to temporarily stop the transfer and call EQ right away.

2.4 若發生 ALARM 時，先按面板上之 [ALARM STOP] 以清除聲響，再按螢幕右下方之 [ALM DSP] 鍵一次，抄錄 ALARM 號碼及訊息，同時在記錄簿備註記錄當時程式執行至第幾步驟，當時之溫度 PV， 並立刻通知設備工程師及製程工程師處理。

When alarm occurs, press the [ALARM STOP] button on the panel to clear off the sound, and press the [ALM DSP] button once to write down the alarm number and message. At the same time, write down in the log book which step the recipe has been run to, the temperature PV at that time, and call PE and EQ right away.

2.5 除 O.I. 提到有關之操作外，其它面板上之按鍵，控制面板螢幕內之程式內容及任何設定參數皆不可擅自更改。

Except for operations mentioned in the O.I., DO NOT touch other buttons on the panel or alter the recipe contents and any set parameters on the screens of the control panel.

2.6 瞬間停電時，立刻檢查面板各項設定 (氣體、溫度、閥) 之狀況，同時在記錄簿上記錄當時之步驟及當時之溫度 PV，並通知製程及設備工程師處理。

When there is instantaneous blackout, check immediately various settings on the panel (gas, temperature and valves) and write down in the log book the step and temperature PV at that time and call PE and EQ right away.

2.7 操作過程中，若不小心按觸到 MODEL 1000 螢幕中之任何非指定之功能鍵，只要按螢幕左下方之 ESC 鍵即可跳出無礙。

If you inadvertently press any non-specified function button on the Model 1000 screen during the operation, just press the ESC key at the lower left of the screen to get out of it.

2.8 F43 M/D 發生在第 19-27 步驟 產品 RWK，F31 M/D 發生在第 20-25 步驟 產品 RWK (附件 2)。

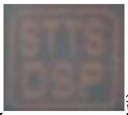
2.9 PM 或 復機時填寫 polyimide 爐管溫度紀錄表 (附件 3)。

3. 〝KOYO 爐管使用記錄表〞填寫時機如下：

3.1 測機時。

3.2 機台發生異常狀況時。如：跳電、當機、ALARM........ 時。

4. 〝KOYO 爐管使用記錄表〞記錄方式 (Record in the KOYO Furnace using table)：

4.1 於反應步驟開始，即 RUN 開始後 2 小時 40 分鐘，按右下方之

鍵，可見反應設定溫度 SP 為 400.0℃，<u>*將反應溫度 PV 之溫度使用 C/P-A系統做參數收集*</u>。

4.1 At the beginning of the reaction step, that is, 2 hours and 40 minutes after the start of RUN, press the button at the bottom right to see the reaction set temperature SP 400.0 ℃,

<u>*Use the C/P-A system to collect the temperature of the reaction temperature PV.*</u>

伍、例行作業 (Routine Operation)

1.每 50 個 RUN，做檔片更換，更換動作如下：

Change the dummy wafers after every 50 runs by the following procedures:

1.1 按螢幕右下方之 W/H DSP 鍵，進入晶舟座轉盤之控制螢幕。

Press the W/H DSP button at the lower right of the screen to enter the control menu of the cassette turntable.

1.2 按螢幕上轉盤之 A 鍵，則可將 A 晶舟朝外，直接取出晶舟更換 25 片新檔片入內，再將晶舟放回轉盤上。

Press the A key on the turntable to move cassette A toward outside. Take out the cassette directly and replace with 25 pieces of new dummy wafers, then put the cassette back in the turntable.

1.3 同理依序更換 B 、 C 、 D 、 E 、 F 晶舟內之檔片。

Follow the same steps to change the dummy wafers in Cassettes B, C, D, E, and F.

2. 產品每 10 個 RUN 後，RUN 程式 6 (O2BAKE) 一次作 BURN OUT 清管，故不需上、下貨，直接進入 PROCESS OPERATION 螢幕執行程式 6 即可，若發現 O2BAKE 遺漏未執行，則立即在該 RUN 結束後，補 RUN 一次 O2BAKE，不累計次數，其餘亦不作更動。

After each 10 runs, run recipe 6 (02BAKE) once to burn out the furnace, thus there is no need to load and unload products. Go instead directly to Process Operation to execute recipe 6. If finding out that O2BAKE execution was missed, do make-up run of O2BAKE once after the current run without adding up this run. The rest is the same as regular run.

3.每班交班後，應做例行清潔檢查，機台周圍及轉盤應保持清潔，若發現晶片碎屑或灰塵雜物,立即通知設備工程師處理。

After handover between shifts, conduct routine cleaning check. The surroundings and turntable of the machine should be kept clean. If wafer chips or dust is found, call EQ right away.

4.烘烤爐管所使用之檔片使用 50 次後報廢，新檔片之來源為每季 TRK 退下之測機控片,使用前需先去光阻.刻號再交 F43.F31 furance 使用。刻號原則為

PI-F43-01 . PI-F43-02 . PI-F43-03 . PI-F43-04 . PI-F43-05 . PI-F43-06

PI-F31-01 . PI-F31-02 . PI-F31-03 . PI-F31-04 . PI-F31-05 . PI-F31-06

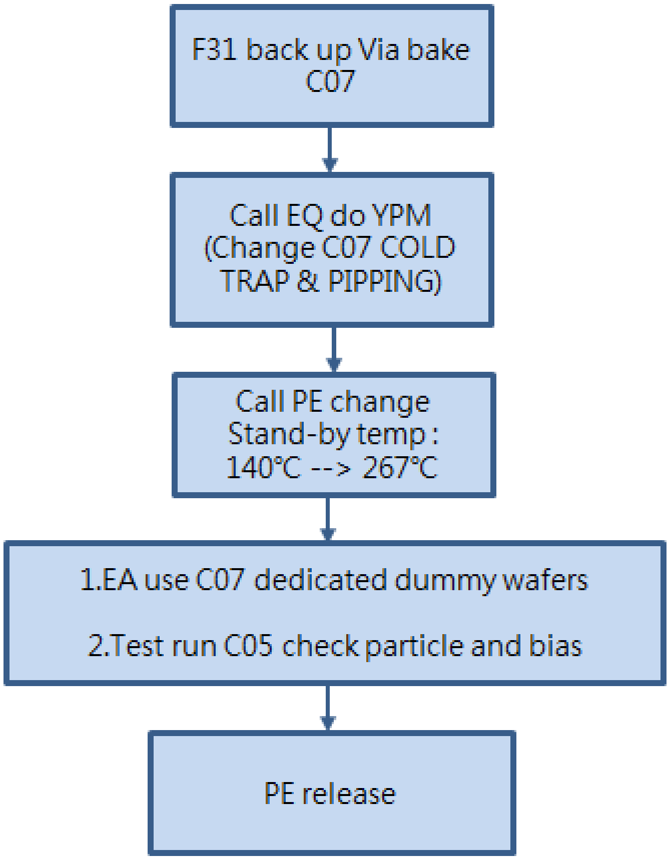
Dummy wafers of bake furnace are scrapped after being used 50 times. The source of new dummy wafers are monitor wafers of TRK.Before using new PI furnace monitor wafer,must do mark by daser-mark. Mark principle is

PI-F43-01 . PI-F43-02 . PI-F43-03 . PI-F43-04 . PI-F43-05 . PI-F43-06

PI-F31-01 . PI-F31-02 . PI-F31-03 . PI-F31-04 . PI-F31-05 . PI-F31-06

陸、F31 back up Via bake C07 操作說明 (F31 back up Via bake C07 instruction)

1.Back up 流程 : (Back up flow)



2.程式內容 : (Recipe contents)

2-1.RECIPE NUMBER：007

RECIPE NAME：F31C07 [F31(B29)]

2-1-1.Stand-by temp : 140℃ --> 267℃

LINE	INST	VALUE 1	VALUE 2
1	CASCT		
2	PIDM	0	
3	T1		
4	TEMP	267	0
5	WAIT	0h0m15s	
6	CTO		
7	CTI		
8	BED		
9	SHC		
10	WAIT	0h0m15s	
11	LD		
12	SHO		
13	BEU		
14	WAIT	0h30m0s	
15	BED		
16	SHC		
17	WAIT	0h10m0s	
18	ULD		
19	CTO		
20	SHO		
21	BEU		
22	END		

2-2.RECIPE NUMBER：009

RECIPE NAME：C05 [F31(B29)]

LINE	INST	VALUE 1	VALUE 2
1	CASCT		
2	PIDM	0	
3	T1		
4	TEMP	267	0
5	WAIT	0h0m15s	
6	CTO		
7	CTI		
8	BED		
9	SHC		
10	WAIT	0h0m15s	
11	LD		
12	SHO		
13	BEU		
14	TEMP	425	3
15	WAIT	0h30m0s	
16	TEMP	267	3
17	BED		
18	SHC		
19	WAIT	0h10m0s	
20	ULD		
21	CTO		
22	SHO		
23	BEU		
24	END		

3.測機操作說明 : (Test run instruction)

3-1.F31 back up C07,機台 PM 後做 test run“C05” 程式並加放 3片 “EDC 控片”及1 片 “PTC 控片” (補滿檔片)。控片置放相關位置如下

Cassette slot	控片擺放種類	參數對應位置
A Cassette Slot 1	EDC 控片	C1
D Cassette Slot 1	EDC 控片	C4
D Cassette Slot 2	PTC 控片	
F Cassette Slot 1	EDC 控片	C6

註 : EDC 厚度控片製作流程為 : 1. PE oxide IM 4.7K → 2. SOG R7 #1

F31 bake up C07 , machine do test run “C05” recipe after PM (fill the boat with dummy wafer)。Put 3 piece of EDC monitor wafer and 1 piece of PTC monitor wafer as below

Cassette slot	The type of monitor	Parameter position
A Cassette Slot 1	EDC monitor wafer	C1
D Cassette Slot 1	EDC monitor wafer	C4
D Cassette Slot 2	PTC monitor wafer	
F Cassette Slot 1	EDC monitor wafer	C6

※ EDC fabrication : 1. PE oxide IM 4.7K ➔ 2. SOG R7 #1

3-2.機台復機之 Test run 結果請依控片位置做轉帳並通知製程工程師檢查 test run 結果

Test run result please follow monitor wafer position then call PE to check test run data.

3-2-1 進爐管前 EDC 厚度轉帳參數 : SOG_C1, SOG_C4, SOG_C6

The parameter of EDC thickness (before move in) : SOG_C1, SOG_C4, SOG_C6

3-2-2 出爐管後 EDC 厚度轉帳參數: CURING_C1, CURING_C4, CURING_C6

The parameter of EDC thickness (after C05 recipe) :

CURING_C1, CURING_C4, CURING_C6

3-2-3 出爐管後 EDC 厚度轉帳參數與進爐管前 EDC 厚度轉帳參數差 :

C05_B15_C1 , C05_B15_C4 , C05_B15_C6 .

ex : “C05_B15_C1” = “CURING_C1” － “SOG_C1”

The parameter of thickness difference is C05_B15_C1 , C05_B15_C4 and C05_B15_C6 .

ex : “C05_B15_C1” = “CURING_C1” － “SOG_C1”

3-2-4 出爐管後 particle 轉帳參數為 : PARTICLE

The parameter of particle is PARTICLE

3-3. F31 back up C07測機必須在規格內才能把機台 release 生產，如不合規格通知製程工程師處理。

規格如下: (Spec. table)

參數 parameter	量測機台 Measure Machine	量測程式 Measure Recipe	Spec. Limit
SOG_CX	NM-3	#1603 W-VIA R7 PEOxide 5pt	9010 +/- 600A
CURING_CX	NM-3	#1605 W-VIA R7 C PEOxide 5pt	8940+/-600A
C05_B15_CX	N/A	N/A	-190 ~ 170A
PARTICLE	SS	BARE-0.2um	<50 ea

About F31 back up C07 , Release the machine must sure that test run data in spec, if not please contact PE to handle it.

3-4.F31 back up C07,機台每月需做例測 C05 程式。測機方式及轉帳方式同 PM 測機。

每班每個月依下表之日期執行測機。

機台 (Machine)	F31
測機程式 Test run Recipe / 班別&日期 Date & Shift	C05
日 (Day)	20
夜 (Night)	5

F31 back up C07, machine must do routine test run. ※ Please refer to PM test run rule .

3-5.測 particle MW 來源\機台測機時宜與程式\量測機台程式:

Test particle monitor wafer source\right time that test and recipe\measure recipe:

3-5-1.Particle MW 請使用 RP-A type MW, particle < 150ea 方可做測機MW

Release to monitor particle wafer for RP-A type MW and particle < 150ea only

3-5-2.量測機台程式: 進前與進後程式皆為BARE-0.2um

Measure recipe: forward or backward data for test particle recipe name is BARE-0.2um

3-5-3.Particle MW < 150ea 可重複使用

It’ s recycled to used the MW when particle < 150ea

3-5-4.Particle MW > 150ea,請集中在回收盒,滿1盒時做particle回收清洗後可重複使用

表單編號:G2920-0009-84

Particle > 150ea, please put away the recycle box. if the box full, please send to do clean

Recycled the MW for test particle item. Data NO: G2920-0009-84

3-5-5.Particle控片不符使用,需要領一盒RP-A型MW,請通知PE工程師

If get the box RP-A type MW, please notice PE engineer

3-6.任何時機 測 Particle OOC/OOS 時需將圖形用影像截取方式存檔，存檔方式請參閱技資編號：G3214-0502。

If particle OOC or OOS, it must be save the particle pattern to computer O.I number: G3214-0502

4.檔片注意事項 : (dummy wafer Notes)

4-1.檔片之來源為OXIDE/SI3N4檔片轉WSIX控片前的wafer。參閱技資編號：G3212-0009-82

The dummy wafer source as OXIDE dummy wafer. Data No：G3212-0009-82

4-2.Back up C07 專用擋片 ，嚴禁Polyimide(F43)使用。

Back up C07 dedicated dummy wafers .Forbidden F43。

柒、品質控制 (Quality Control)

1.爐管 THERMAL COUPLE，設備工程師每一年定期檢驗一次，檢驗結果及方式詳見儀校中心之記錄。

The thermal couple of the furnace should be checked periodically once a year by EQ. See the records of Calibration Center for check results and method.

2.為求溫度之 OVER SHOTTING 不超過 7℃，每月檢查溫度之 ON LINE DATA，以確保溫度控制之穩定，結果登記在 KOYO 爐管使用記錄表，每月最後一天 (或次日)之備註欄。

To make sure the over shotting of temperature does not exceed 7℃, check the on line data of temperature every month to make sure the temperature is kept stable. Write down the results in remark column of the KOYO Furnace Record List on the last day of the month (or next day)

3.爐管 PM 時，設備工程師每一年定期檢驗一次，檢驗結果及方式詳見儀校中心之記錄。

捌、控片使用及回收流程 (Use and Recycle of Dummy Wafers)

參閱 "FAB 擴散區控片使用及回收流程規範"，技資編號：G2920-0009。

See "FAB Diffusion Area Monitor Wafer Use and Recycle Flow Process Rules", Doc. No.: G2920-0009.

玖、附件一 (Attachment)

KOYO 爐管使用記錄表/ KOYO furnace using table

Date	input：	output：	Date	input：	output：
Time	input：	output：	Time	input：	output：
Work No.	input：	output：	Work No.	input：	output：
accumulative runs			accumulative runs		
Pcs Product Lot No.	A B C D E F		Pcs Product Lot No.	A B C D E F	
recipe			recipe		
processing temperature	TOP CENT BTM . . .		processing temperature	TOP CENT BTM . . .	
record time	(10)		record time		
N2 flow rate	(11)		N2 flow rate		
O2 flow rate	(12)		O2 flow rate		
comment	(13)		comment		

Date	input：	output：	Date	input：	output：
Time	input：	output：	Time	input：	output：
Work No.	input：	output：	Work No.	input：	output：
accumulative runs			accumulative runs		
Pcs Product Lot No.	A B C D E F		Pcs Product Lot No.	A B C D E F	
recipe			recipe		
processing temperature	TOP CENT BTM . . .		processing temperature	TOP CENT BTM . . .	
record time			record time		
N2 flow rate			N2 flow rate		
O2 flow rate			O2 flow rate		
comment			comment		

表單編號：G3212-2402-01-C

附件(一～1)：

KOYO 爐管使用記錄表/ KOYO furnace using table

Date	input：(1)	output：	Date	input：	output：
Time	input：(2)	output：	Time	input：	output：
Work No.	input：(3)	output：	Work No.	input：	output：
accumulative runs	(4)		accumulative runs		
Pcs Product Lot No.	A B C D E F .(5) . . . . . .(6) . . . . . .(7)		Pcs Product Lot No.	A B C D E F	
recipe	(8)		recipe		
processing temperature	TOP CENT BTM (9) . . .		processing temperature	TOP CENT BTM . . .	
record time	(10)		record time		
N2 flow rate	(11)		N2 flow rate		
O2 flow rate	(12)		O2 flow rate		
comment	(13)		comment		

Date	input：	output：	Date	input：	output：
Time	input：	output：	Time	input：	output：
Work No.	input：	output：	Work No.	input：	output：
accumulative runs			accumulative runs		
Pcs Product Lot No.	A B C D E F		Pcs Product Lot No.	A B C D E F	
recipe			recipe		
processing temperature	TOP CENT BTM . . .		processing temperature	TOP CENT BTM . . .	
record time			record time		
N2 flow rate			N2 flow rate		
O2 flow rate			O2 flow rate		
comment			comment		

表單編號：G3212-2402-01-C

附件(一～2)：

表格填寫說明/Table write down describe

表格名稱/Table name：KOYO 爐管使用記錄表/KOYO furnace using table

項目	欄位名稱	填寫方式及內容描述
1	Date 日期	write down run start and edd date 填寫產品 RUN 貨開始及結束的日期
2	Time 時間	write down run start and end time 填寫產品 RUN 貨開始及結束的時間
3	Work NO. 工號	write down operater`s work NO. 填寫操作者的工號
4	accumulative runs 累積次數	write down accumulated runs,after 49 runs reaccumulated 填寫使用次數的累加，滿 49 次歸 0 重新累計
5	pcs 片數	write down pcs,product NO.,lot NO. by each caseaed. 依據每個晶舟內之產品，分別填寫其片數、型號、批號
6	product 型號	
7	lot NO. 批號	
8	recipe 程式	write down processe recipe 填寫執行的程式名稱
9	processing temperature 反應溫度	write down temperature in man process step 填寫程式執行當中，主要反應步驟時的溫度
10	record time 抄錄時間	write down record time 填寫抄錄時間
11	N2 flow rate N2 流量	write down N2 flow rate in mam process step 填寫程式執行當中，主要反應步驟時的 N2 流量
12	O2 flow rate O2 流量	run recipe #6,burn out furnace,write down O2 flow rate in man process step(750℃) RUN 程式 6，作 BURN OUT 清管時，記錄主要反應步驟(750℃)之 O2 流量
13	comment 備註	write down PE required iteam 可記錄其他製程要求的項目

附件二

POLYIMIDE 烘烤不良 RWK 單(REWORK2F)

ROUTE：____________型 號：__________批 號：________

層 次：__________________RWK 次數：________

原 因：____________________製程簽名：__________

□ 整批 □ 非整批，刻號：______________片數：________

代號	程 序	站別	工 號	日 期	進 量	出 量	時 間	機 台
960	光阻去除	C						
請送至 A-TUNNEL								
961	電 漿 清 洗	A						PSC
注意：POLYIMIDE 去除於換酸前再作								
962	POLYIMIDE 去除	A						SOL
963	A E I	A						AEI-
	是否洽PE	□ 否		□ 是		製程:__________		
請送回黃光區Return Operation：								
	POLYIMIDE 覆蓋	C						TRK-
	光 阻 覆 蓋	F						TRK-
	Buffer COAT 對準	F						ASM-
	曝光量： mj	MASK：						
	正 顯 影	F						TRK-
	A D I	F						ADI-
	光 阻 去 除	C						TRK-
請送至 B-TUNNEL								
	POLYIMIDE 烘烤	B						F-

備註：

是否回洽PE

□是 ，刻號 _______________

□否 ，下接 _______________ 站

幹部：

表單編號：G3212-2402-02-A

附件三

polyimide爐管溫度紀錄表

Time	溫度	TOP	CNT	BTM	Time	溫度	TOP	CNT	BTM
	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			
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	SP					SP			
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	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			
	SP					SP			
	PV					PV			

表單編號：G3212-2402-03-A

拾、生效與修訂 (Documentation effectiveness and revision)

本規範之公佈實施及其修訂核准層級皆依會簽/核決/分發依循範例為之。